

Francesco Maria Vinciguerra

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🏠 Milan, Italy & Lausanne, Switzerland

Research Experience

Visiting Researcher — AI Safety UNIVERSITY OF WARWICK — PROF. PAUL JENKINS 2026

- Developing rare-event sampling methods for language models and masked diffusion models, with applications to discovering low-probability harmful behaviours and estimating their occurrence probabilities.
- Studying the theoretical and empirical properties of the resulting estimators, including bias, variance, convergence, and computational efficiency; submitted to NeurIPS 2026 workshop on AI safety.

Research Project in Stochastic Analysis EPFL — PROF. MARTIN HAIRER 2026

- Developed a proof strategy for a functional limit theorem for the Lorentz Mirror Walk.

Industry Experience

Data & AI Intern — AI Refinery ACCENTURE, MILAN Jun 2025 – Sep 2025

- Benchmarked LLMs and built evaluation pipelines to identify hallucination, reasoning, and reliability failures.
- Designed multi-agent workflows with model routing, specialized prompting, and RAG to improve grounding and task performance.
- Contributed to the AI Refinery framework by implementing agent-orchestration components and analyzing system-level performance across different configurations.

Education

EPFL — MSc in Applied Mathematics LAUSANNE, SWITZERLAND 2025 – Present **GPA: 5.2/6.**

Relevant coursework: *Stochastic Simulation, Statistical Inference, Deep Learning, Large Deviations, Optimal Transport.*

Bocconi University — BSc in Mathematical and Computing Sciences for AI MILAN, ITALY 2022 – 2025 **GPA: 28/30.** Thesis: *Geometric Group Theory.*

Relevant coursework: *Probability, Bayesian Statistics, Numerical Methods, Algorithms.*

Technical Skills & Awards

Programming & Tools: Python, PyTorch, NumPy/SciPy, Git, Linux, HPC/SLURM.

ML & Methods: Hugging Face, LLM Evaluation, Masked Diffusion Models, Agentic Systems, RAG.

Awards: IMC Trading Challenge — 1st Place (EPFL Forum 2025 Edition).

Languages: Italian (Native), English (C1).